

CURRICULUM VITAE

Shaojie Li

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EDUCATION

- May 2023-May 2025** **Master of Information Technology (Artificial Intelligence Stream; two-year programme)**
University of New South Wales, Sydney, NSW, Australia
Overall grade: Distinction
- Jun 2017-Jun 2021** **Bachelor of Engineering in Internet of Things Engineering**
Shenzhen University, Shenzhen, Guangdong, China

RESEARCH PROFILE

- Trustworthy visual and multimodal AI: uncertainty calibration, failure diagnosis, selective prediction, and risk-aware decisions under distribution shift.
- Reliability-aware medical imaging and image-guided robotics: semantic segmentation, multimodal registration, geometric quality control, and validation before robotic action.
- Reproducible physical-AI experimentation: closed-loop evaluation, low-SNR sensing, embedded inference, ablation studies, and auditable research software.

MANUSCRIPTS UNDER REVIEW

[1] Shaoming Li, **Shaojie Li**, and Fangxun Zhong*. "Diagnosing Candidate Misranking in Learned 3D Dynamics Models via Risk-Averse Closed-Loop Probing." Manuscript under review at the NeurIPS 2026 Evaluations and Datasets Track.

Contribution: Implemented replay-time evaluation and attribution code for multi-seed sweeps, shared-pool dual-oracle analysis, planner/optimizer ablations, risk-gate calibration, and reproducibility; across 32 surgical replay cases, proxy-safe alternatives existed in every 2,048-candidate pool but never entered the model's top-256 rankings, isolating misranking from candidate scarcity.

[2] Shaoming Li, **Shaojie Li**, Che Cheng, and Fangxun Zhong*. "Fail-Closed Validity Contracts for Reliable Robot-Learning Evaluation." Manuscript submitted to the AAAI 2027 Main Technical Track.

Contribution: Audited and reproduced the Python fail-closed validation workflow, verifying schema integrity, source-group role isolation, SHA-256 evidence binding, and pre-learning interception of all 135 registered invalid configurations in the frozen LapGym Reach evaluation setting.

**Corresponding author.*

RESEARCH EXPERIENCE

May 2025-Present **Research Assistant, The Chinese University of Hong Kong**
Supervisor: Professor Yunhui Liu

Project 1: MRI-TRUS Image Processing and Robot Control for Ultrasound-Guided Prostate Biopsy

- Developed Python control prototypes for an Inovance IR-U8-series collaborative robot over TCP and a Beckhoff ADS puncture stage, integrating motion commands, position/state polling, target and timeout register writes, tolerance-based stop logic, and error/status feedback; built a unified Electron/React operator interface.
- Designed and executed an optical-tracking RCM repeatability study across 0, 3.5, 5, and 8 N loads; analyzed 197 quality-controlled line observations and obtained a report-defined pooled repeatability range (RP_all) of 0.53-0.87 mm.
- Built an offline prostate-MRI segmentation and MRI-TRUS registration research-software pipeline using MONAI pretrained models and ITKElastix, implementing geometry validation, 0.5-mm preprocessing, sliding-window inference, native-space restoration, signed-distance staged registration, Dice/HD95/TRE and Jacobian-based regional deformation checks, quality control, provenance tracking, and Python/CLI/local-HTTP interfaces.
- On a frozen 40-case PROMISE12 evaluation set, the segmentation pipeline improved mean Dice from 0.77 to 0.85 and reduced HD95 from 18.65 to 5.31 mm versus the MONAI whole-volume baseline. Evaluated the MRI-TRUS registration workflow across nine mu-RegPro pairs from five MRI groups using contour Dice, HD95, centroid TRE, and Jacobian-based deformation checks.

Project 2: Low-SNR Sensor Time-Series Parameter Estimation and Reliability Modelling

- Developed signal-quality screening, convolutional/recurrent/attention baselines, multi-task objectives, and physics-constrained decoding under a fixed five-fold, record-level OOF protocol, with sub-batch diagnostics for distribution-shift and reliability analysis.
- Built selective-prediction analyses and an int8 native-C inference prototype to quantify risk-coverage trade-offs; verified host-side numerical consistency and embedded-target build compatibility.

Project 3: Surgical Scene Semantic Segmentation and Auditable Model Adaptation (MediSegMan)

- Built and independently executed an auditable offline surgical-scene segmentation pipeline across five public datasets; adapted the public SegMAN implementation and evaluated five public comparison pipelines under dataset-specific protocols. Completed 17 frozen benchmark units covering dataset/label conversion, video-/case-level split checks, validation-based checkpoint selection, grouped metrics, and result provenance.
- Integrated CoordAtt/ECA at the P5/P11 SegMAN decoder stages and evaluated a role-grouped residual-logit recalibration variant. Ran training-side comparisons using class-weighted cross-entropy, Dice, OHEM, and AdapterSIS teacher-logit distillation, and built all-class, foreground, per-class, and role-group metrics with video-/case-level paired bootstrap analyses.
- Completed CaDIS Task I data conversion for 25 videos/4,670 total frames, a preregistered video-level split, three-method training, validation-based checkpoint freezing, and sealed testing. Independently recomputed confusion matrices and verified SHA-256 prediction hashes; across SegMAN, A6, and A6+TAB, sealed-test mIoU ranged from 86.65% to 86.93%.

Feb 2026-Aug 2026 **Research Collaborator | Independent Research Collaboration**

Self-initiated research collaboration with Shaoming Li, PhD Student, The Chinese University of Hong Kong, Shenzhen

Projects: Closed-Loop Reliability and Fail-Closed Evaluation for Robot Learning

- Output: Two second-author manuscripts: one under review at the NeurIPS 2026 Evaluations and Datasets Track and one submitted to the AAAI 2027 Main Technical Track.
- Implemented and extended replay-time evaluation and attribution code for multi-seed sweeps, shared-pool dual-oracle analysis, planner/optimizer ablations, risk-gate calibration, and reproducibility; across 32 surgical replay cases, proxy-safe alternatives existed in every 2,048-candidate pool but never entered the model's top-256 rankings, isolating misranking from candidate scarcity.
- Audited and reproduced the Python fail-closed validation workflow; in the frozen LapGym Reach evaluation setting, verified that all 135 registered invalid configurations were intercepted before learning.

Nov 2021-Apr 2023 **Assistant Engineer, The Innovation Research Institute, The Chinese University of Hong Kong**

Supervisor: Professor Yunhui Liu

Project: Vision-Based Battery-Weld Inspection with Line-Laser Sensing

- Integrated line-laser imaging, photoelectric-gate triggering, conveyor control, and image/point-cloud processing into a battery-weld inspection workflow.
- Connected capture, processing services, APIs, database/storage functions, and deployment components; the project-level end-to-end inspection time decreased from approximately 120 to 27 seconds per test.

Sep 2023-Dec 2023 **COMP9444 Artificial Intelligence Project, University of New South Wales**

Supervisors: Dr Alan Blair and Dr Sonit Singh

Project 1: Real-Time Face Detection and Recognition Using YOLOv5

- Built a YOLOv5-based face-detection and recognition pipeline and compared its speed and detection behaviour with public DSFD and RetinaFace baselines under varied imaging conditions.
- Implemented batch recognition and result tagging for multi-image evaluation and error analysis.

Project 2: HalfCheetah Reinforcement Learning

- Implemented and tuned a Truncated Quantile Critics agent for HalfCheetah, analyzing how reward design and training parameters affected return, locomotion stability, and control behaviour.

May 2024-Aug 2024 **COMP9491 Applied Artificial Intelligence Project, University of New South Wales**

Supervisors: Associate Professor Yang Song and Professor Maurice Pagnucco

Project: Toxic-Comment Severity Rating

- Implemented GPT-based toxicity scoring and compared model outputs under a shared evaluation protocol using the Jigsaw Toxic Severity Rating data.

Nov 2020-May 2021 **BEng Project, Shenzhen University**

Supervisor: Associate Professor Lin Lin, Shenzhen Technology University

Project: Remote Unattended-Terminal Network Fault Recovery

- Designed and built a solo terminal-recovery prototype using a development board, SIM800L cellular communication, and remote reset/network-monitoring logic for unattended fault recovery.
- Tested Wi-Fi control, cellular-message commands, remote reset, and terminal power switching through the project demonstration.

SKILLS

- Programming and AI: Python, C/C++, JavaScript, SQL, PyTorch, MONAI, SimpleITK, ITKElastix, and YOLOv5; segmentation, registration, reinforcement learning, NLP, and 3D point clouds; English: IELTS 7.0
- Systems and methods: Linux/Windows, robot TCP, Beckhoff ADS, Electron/React, and native-C inference; grouped/OOF validation, calibration, selective prediction, ablations, failure analysis, and reproducibility